



0-2-4-5-3-1-0

Don't Care

next

T_0

$D_2 D_1 D_0$

$D_2 D_1 D_0$

$T_2 T_1 T_0$

0 0 0

0 1 0

0 1

0 0 1

0 0 0

0 0

0 1 0

1 0 0

1 1

0 1 1

0 0 1

0 1

1 0 0

1 0 1

0 0

1 0 1

0 1 1

1 1

1 1 0

X X X

X X

1 1 1

X X X

X X

T_2

D_2

$D_1 D_0$

	00	01	11	10
0	0	0	0	1
1	0	1	X	X

$$T_2 = D_2 D_0 + D_1 \overline{D_0}$$

T_1

D_2

$D_1 D_0$

	00	01	11	10
0	1	0	1	1
1	0	1	X	X

$$T_1 = D_1 + \overline{D_2} \overline{D_0} + D_2 D_0$$

T_0

D_2

$D_1 D_0$

	00	01	11	10
0	0	1	0	0
1	1	0	X	X

$$T_0 = \overline{Q_2} \overline{Q_1} Q_0 + Q_2 Q_0$$

Q2 $\rightarrow$ 3 bit updown counter with -ve edge FF

	prev			next					
up/down	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_{Q_2}	T_{Q_1}	T_{Q_0}
0	0	0	0	0	0	1	0	0	1
0	0	0	1	0	1	0	0	1	1
0	0	1	0	0	1	1	0	0	1
0	0	1	1	1	0	0	1	1	1
0	1	0	0	1	0	1	0	0	1
0	1	0	1	1	1	0	0	1	1
0	1	1	0	1	1	1	0	0	1
0	1	1	1	0	0	0	1	1	1
1	0	0	0	1	1	1	1	1	1
1	0	0	1	0	0	0	0	0	1
1	0	1	0	0	0	1	0	1	1
1	0	1	1	0	1	0	0	0	1
1	1	0	0	0	1	1	1	1	1
1	1	0	1	1	0	0	0	0	1
1	1	1	0	1	0	1	0	1	1
1	1	1	1	0	0	0	0	0	1

$T_0 = 1$

T_2	$Q_2 Q_1$			
	00	01	10	11
up Q_3	00	0	0	1
01	0	0	1	0
11	1	0	0	0
10	1	0	0	0

$$T_2 = d_2 Q_1 + Q_2' Q_1' M$$

$$T_2 = M'$$

T	$Q_2 Q_1$			
	00	01	10	11
up Q_3	00	0	1	1
01	0	1	1	0
11	1	0	0	1
10	1	0	0	1

$$T_1 = M' Q_1 + M Q_1'$$

$$T_1 = M' Q_1 + M Q_1'$$

Q3, Assume 000 initial q_0, q_1, q_2

$1 \rightarrow 2 \rightarrow 7 \rightarrow 0 \rightarrow 1$